

Convection HT

Rate of mass diffusion $\dot{m}$

$$\dot{m}_1 = -\rho D_{12} \nabla w_1$$

defined as Fick's Law.

$$\dot{m}_2 = -\rho D_{21} \nabla w_2$$

where D_{12} is the binary diffusion coefficient

$$D_{12} = D_{21} = D$$

If we add the mass concentrations $w_1 + w_2 = 1$

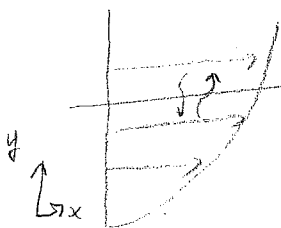
$$\dot{m}_1 + \dot{m}_2 = -\rho D \nabla(1) = 0$$

If expression is non zero, mass will accumulate !!

Other effects on $\vec{g}$ and $\dot{m}$ is discussed in Appendix A

Shear Stresses

We have regions of high & low velocity.



stress occurs when particles move by diffusion because the particles bring with them momentum.
 τ_{yx} is stress on y surface in the x dir

$$\vec{\tau}_{yx} = \mu \frac{\partial u}{\partial y}$$

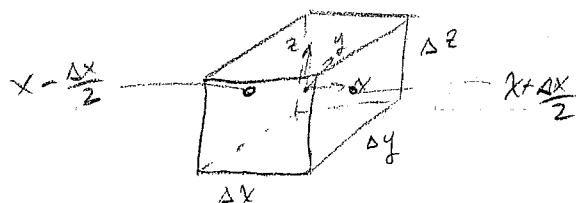
↑ direction
surface

Conservation Equations

$$\vec{v}(\vec{x}, t) = \hat{i} u(x, y, z, t) + \hat{j} v(x, y, z, t) + \hat{k} w(x, y, z, t)$$

same thing with $T(\vec{x}, t)$, $\rho(\vec{x}, t)$, $\rho(\vec{x}, t)$, $\rho(\vec{x}, t)$, $\rho(\vec{x}, t)$ concentration
All these need to be found.

Mass



$$u(x + \frac{\Delta x}{2}, y, z, t) = u(x, y, z, t) + \frac{\partial u}{\partial x} \bigg|_{x, y, z} \frac{\Delta x}{2} + O(\Delta x^2)$$

$$- \dot{m}_{\text{stored}} = \sum \dot{m}_{\text{in}} - \sum \dot{m}_{\text{out}}$$

$$(\dot{m}_{\text{in}} - \dot{m}_{\text{out}})_x = \rho u \bigg|_{x = \frac{\Delta x}{2}} \Delta y \Delta z - \rho u \bigg|_{x = -\frac{\Delta x}{2}} \Delta y \Delta z$$

If the face is small enough, ρ is same along the entire face.

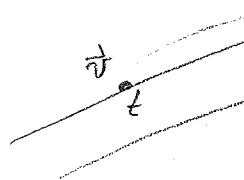
$$(\dot{m}_{in} - \dot{m}_{out})_x = \Delta y \Delta z \left[\rho u + \frac{\partial(\rho u)}{\partial x} \frac{\Delta x}{2} - \rho u + \frac{\partial(\rho u)}{\partial x} \frac{\Delta x}{2} \right]$$

$$\Delta V = \Delta x \Delta y \Delta z$$

$$- \left[\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} \right] \Delta V = - \frac{\partial \rho}{\partial t} \Delta V$$

$$\boxed{\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0}$$

Substantial Derivative (Total, Material, Eulerian)...

 $Q(x, y, z, t)$ where Q is some field quantity.

What is the change of Q at $t + \delta t$.

The coordinate of pt after $t + \delta t$ is $(\vec{x} + \vec{v} \delta t)$

We can evaluate new value of Q as

$$Q(x + u \delta t, y + v \delta t, z + w \delta t, t + \delta t)$$

$$\frac{DQ}{Dt} = \lim_{\delta t \rightarrow 0} \frac{Q(\vec{x} + \vec{v} \delta t, t + \delta t) - Q(\vec{x}, t)}{\delta t}$$

$$\begin{aligned} \text{Since } Q(x + u \delta t, y + v \delta t, z + w \delta t, t + \delta t) &= Q(x, y, z, t) + \frac{\partial Q}{\partial x} u \delta t \\ &+ \frac{\partial Q}{\partial y} v \delta t \\ &+ \frac{\partial Q}{\partial z} w \delta t \\ &+ \frac{\partial Q}{\partial t} \delta t \end{aligned}$$

$$\frac{DQ}{Dt} = \frac{\partial Q}{\partial t} + u \frac{\partial Q}{\partial x} + v \frac{\partial Q}{\partial y} + w \frac{\partial Q}{\partial z}$$

$$\boxed{\frac{DQ}{Dt} = \frac{\partial Q}{\partial t} + (\vec{v} \cdot \nabla) Q}$$

where $\nabla = i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z}$

$$\vec{v} = i u + j v + k w$$

$$\vec{v} \cdot \nabla = u \frac{\partial}{\partial x} + v \frac{\partial}{\partial y} + w \frac{\partial}{\partial z}$$

Using the chain rule,

$$\nabla \cdot (f \vec{v}) = f \nabla \cdot \vec{v} + \vec{v} \cdot \nabla f$$

Since

$$\frac{Df}{Dt} = \frac{\partial f}{\partial t} + \vec{v} \cdot \nabla f$$

The continuity eqn becomes

$$\boxed{\frac{Df}{Dt} + f \nabla \cdot \vec{v} = 0}$$

Case f is constant.

$$\boxed{\nabla \cdot \vec{v} = 0}$$

case Incompressible Flow. $\frac{D\rho}{Dt} = 0$

We can have particles with different density !!

But the density of each particle is zero.

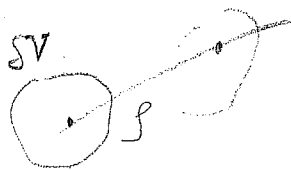
ex. Atmosphere is stratified.

$$\nabla \cdot \vec{V} = 0$$

case Steady Flow $\frac{\partial}{\partial t} = 0$

$$\nabla \cdot (\rho \vec{V}) = 0$$

An Alternative Approach



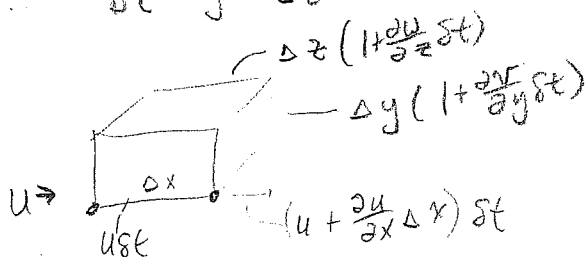
$$\delta m = \rho \delta V$$

$$\frac{D\delta m}{Dt} = 0$$

The change in the mass moving around is zero.

$$\frac{D}{Dt}(\rho \delta V) = 0$$

$$\delta V \frac{D\rho}{Dt} + \rho \frac{D\delta V}{Dt} = 0$$



So the amount moved is $\Delta x (1 + \frac{\partial u}{\partial x} \delta t)$

$$\begin{aligned}\text{new volume} &= \delta V \left(1 + \frac{\partial u}{\partial x} \delta t\right) \left(1 + \frac{\partial v}{\partial y} \delta t\right) \left(1 + \frac{\partial w}{\partial z} \delta t\right) \\ &= \delta V [1 + \nabla \cdot \vec{v} \delta t + \dots]\end{aligned}$$

$$\frac{D\delta V}{Dt} = \frac{\delta V \nabla \cdot \vec{v} \delta t}{\delta t}$$

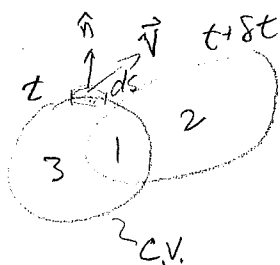
$$\frac{D\delta V}{Dt} = \delta V \nabla \cdot \vec{v} \quad \text{--- (2)}$$

The dilatation $\nabla \cdot \vec{v} = \frac{1}{\delta V} \frac{D\delta V}{Dt}$, i.e. divergence of the velocity field.

If $\nabla \cdot \vec{v} = 0$, volume change is zero.

Inserting (2) into (1),

$$\boxed{\frac{D\rho}{Dt} + \rho \nabla \cdot \vec{v} = 0}$$



$$m_1(t) + m_2(t) = m_1(t + \delta t) + m_2(t + \delta t)$$

$$m_1(t + \delta t) - m_1(t) = m_3(t) - m_2(t + \delta t)$$

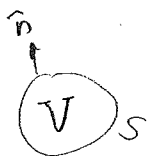
$$\frac{m_1(t + \delta t) - m_1(t)}{\delta t} = \frac{m_3(t) - m_2(t + \delta t)}{\delta t}$$

Let $\delta t \rightarrow 0$,

$$\frac{\partial}{\partial t} \int_{C.V.} \rho dV = - \int_{C.S.} \rho \vec{v} \cdot \hat{n} ds$$

net mass flowing out.

Proof of divergence theorem



$$\int_V \nabla \cdot \vec{g} dV = \int_S \vec{g} \cdot \hat{n} ds$$

We need g to be differentiable & differential needs to be continuous.

Using divergence theorem,

$$\frac{\partial}{\partial t} \int_V \rho dV = - \int_V \nabla \cdot \rho \vec{v} dV = - \int_S \rho \vec{v} \cdot \vec{n} dS$$

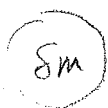
$$\int_V \left(\frac{\partial \rho}{\partial t} + \nabla \cdot \rho \vec{v} \right) dV = 0$$

Shrinking dV ,

$$\boxed{\frac{\partial \rho}{\partial t} + \nabla \cdot \rho \vec{v} = 0}$$

Momentum Equation

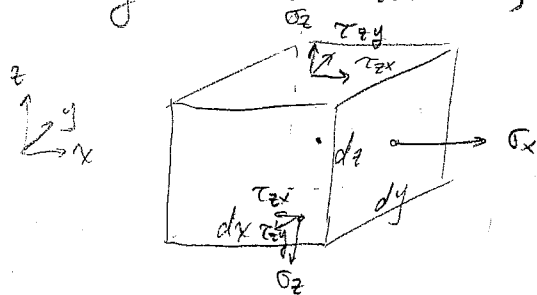
A fluid element



$$\vec{\text{Force}} = \frac{D}{Dt} (Sm \vec{v}), \quad \frac{D(Sm)}{Dt} = 0$$

$$\vec{\text{Force}} = \rho \delta V \frac{D \vec{v}}{Dt}$$

Looking at an element,



A body force acts on the body from a distance.

$$\vec{f} = f\vec{j} = (X, Y, Z)$$

Inserting into Force eqn,

$$\delta V \vec{f} + \{\text{Surface Forces}\} = \rho \delta V \frac{D\vec{v}}{Dt}$$

Surface force on x-dir

$$\begin{aligned} \text{Surface Force}_x &= dydz \left[\sigma_x + \frac{\partial \sigma_x}{\partial x} \frac{dx}{2} - \left(\sigma_x - \frac{\partial \sigma_x}{\partial x} \frac{dx}{2} \right) \right] + \\ &\quad dx dy \left[\tau_{zx} + \frac{\partial \tau_{zx}}{\partial z} \frac{dz}{2} - \left(\tau_{zx} - \frac{\partial \tau_{zx}}{\partial z} \frac{dz}{2} \right) \right] + \\ &\quad dx dz \left[\tau_{yx} + \frac{\partial \tau_{yx}}{\partial y} \frac{dy}{2} - \left(\tau_{yx} - \frac{\partial \tau_{yx}}{\partial y} \frac{dy}{2} \right) \right] \\ &= \delta V \left[\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} \right] \end{aligned}$$

Working out y & z, we get

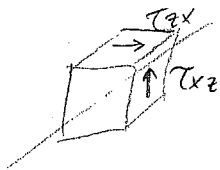
$$\left\{ \begin{aligned} \rho \frac{Du}{Dt} &= \rho X + \frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} \\ \rho \frac{Dv}{Dt} &= \rho Y + \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} + \frac{\partial \tau_{zy}}{\partial z} \\ \rho \frac{Dw}{Dt} &= \rho Z + \frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_z}{\partial z} \end{aligned} \right\} \quad \text{the momentum eqn.}$$

For fluids without no molecular distribution of moments,

$$\begin{aligned} \tau_{yz} &= \tau_{zy} \\ \tau_{xy} &= \tau_{yx} \\ \tau_{xz} &= \tau_{zx} \end{aligned}$$

Proof

Look at rotation around y-axis. τ_{xz} and τ_{zx} contributes to rotation.



$$\tau_{xz} dy dz dx - \tau_{zx} (dx dy) dz = dV (\tau_{xz} - \tau_{zx}) = dI_y \omega_y$$

$$\text{Since } dI_y = dV \cdot dl^2$$

$$dV (\tau_{xz} - \tau_{zx}) = dV dl^2$$

Since dl^2 is small, $\tau_{xz} = \tau_{zx}$.

$$T = \begin{bmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_y & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z \end{bmatrix}$$

For a Newtonian, isotropic fluid,

$$\tau_{xy} = \tau_{yx} = \mu \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right)$$

$$\tau_{yz} = \tau_{zy} = \mu \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right)$$

$$\tau_{xz} = \tau_{zx} = \mu \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right)$$

$$\sigma_x = -p - \frac{2}{3} \mu \nabla \cdot \vec{v} + 2\mu \frac{\partial u}{\partial x}$$

$$\sigma_y = -p - \frac{2}{3} \mu \nabla \cdot \vec{v} + 2\mu \frac{\partial v}{\partial y}$$

$$\sigma_z = -p - \frac{2}{3} \mu \nabla \cdot \vec{v} + 2\mu \frac{\partial w}{\partial z}$$

the constitutive equations. Putting these into the momentum eqns,

$$\begin{aligned} \rho \frac{Du}{Dt} = \rho X - \frac{\partial p}{\partial x} + \frac{\partial}{\partial x} \left[\mu \left(\frac{\partial u}{\partial x} - \frac{2}{3} \nabla \cdot \vec{v} \right) \right] + \frac{\partial}{\partial y} \left[\mu \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) \right] \\ + \frac{\partial}{\partial z} \left[\mu \left(\frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} \right) \right] + \frac{\partial p}{\partial x} \end{aligned}$$

and so on with y & z momentum.

The problems we will consider will have constant properties. $\rho = \text{const}$

$$\mu = \text{const}$$

$$\nabla \cdot \vec{v} = 0$$

The momentum becomes

$$\left. \begin{aligned} \int \frac{D\vec{v}}{Dt} &= \vec{f} - \nabla p + \mu \nabla^2 \vec{v} \\ \nabla \cdot \vec{v} &= 0 \end{aligned} \right\}$$

Energy Equation

Forms of energy per unit mass.

- internal thermal energy: e , specific internal energy: E
- KE $v^2 = \vec{v} \cdot \vec{v}$: $\frac{v^2}{2}$
- PE : gz
- enthalpy h : $h = e + p/\rho = e + pV$

Consider a fluid element

$$\rho dV$$

1st Law of thermo : $\dot{Q} = \frac{DE}{Dt} + \dot{W}$

where

$$\frac{DE}{Dt} = \rho dV \left\{ \frac{De}{Dt} + \frac{1}{2} \frac{D}{Dt} (u^2 + v^2 + w^2) \right\}$$

Look at element,



$$\vec{q} = -k \nabla T$$

$$-k \frac{\partial T}{\partial x} + \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) \left(-\frac{dx}{2} \right) dx dy dz - k \frac{\partial T}{\partial x} + \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) \left(\frac{dx}{2} \right) dx dy dz$$

$$\dot{Q}_x = dydzdx \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \dots \quad (\text{in-out})$$

$$\dot{Q} = \delta V \nabla \cdot k \nabla T + g''' \delta V$$

where g''' is internal heating per unit volume.

Mass Conservation:

$$\frac{D\rho}{Dt} + \rho \nabla \cdot \underline{v} = 0, \quad \textcircled{+} \text{ Continuity}$$

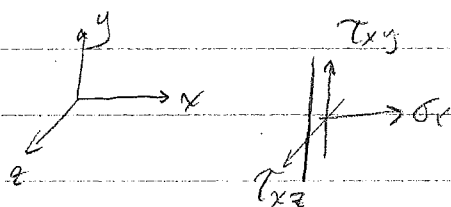
$$\frac{D}{Dt} = \frac{\partial}{\partial t} + \underline{v} \cdot \nabla$$

$$\underline{v} \cdot \nabla = u \frac{\partial}{\partial x} + v \frac{\partial}{\partial y} + w \frac{\partial}{\partial z}$$

Momentum

Tensor: $\underline{T} = T_{ij}$

$$\underline{T} = \begin{bmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_y & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z \end{bmatrix} \quad \text{stress tensor}$$



From angular momentum conservation: $T_{ij} = T_{ji}$

$$\rho \frac{D\underline{x}}{Dt} = \underline{F} - \nabla p + \nabla \cdot \left[\mu (\nabla \underline{x} + \nabla \underline{x}^T) - \frac{2}{3} \nabla (\nabla \cdot \underline{x}) \right]$$

or writing the momentum in index notation,

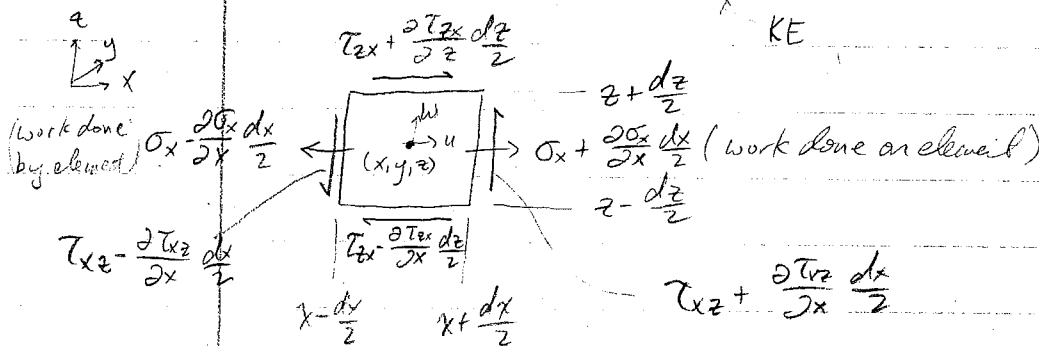
$$\rho \frac{Dv_i}{Dt} = f_i - \frac{\partial p}{\partial x_i} + \sum_{j=1}^3 \frac{\partial}{\partial x_j} \left[\mu \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} \right) - \frac{2}{3} \delta_{ij} (\nabla \cdot \underline{v}) \right]$$

Energy Equation

$$\dot{Q} = \frac{DE}{Dt} + \dot{W}$$

$$\dot{Q} = \{ \nabla \cdot (k \nabla T) + \rho''' \} \delta V, \quad q = -k \nabla T$$

$$\frac{DE}{Dt} = \int dV \left\{ \frac{De}{Dt} + \frac{1}{2} \frac{Dv^2}{Dt} \right\}$$



The work done on all the surfaces in x-dir:

for u & velocity same dir $\Rightarrow$ work done to element $\Rightarrow (-)$

$$\dot{W}_{\sigma_x} = -dzdy \left\{ \left(\sigma_x + \frac{\partial \sigma_x}{\partial x} \frac{dx}{2} \right) \left(u + \frac{\partial u}{\partial x} \frac{dx}{2} \right) - \left(\sigma_x - \frac{\partial \sigma_x}{\partial x} \frac{dx}{2} \right) \left(u - \frac{\partial u}{\partial x} \frac{dx}{2} \right) \right\}$$

$$\dot{W}_{\sigma_x} = -dV \frac{\partial (u \sigma_x)}{\partial x}$$

$$\dot{W}_{\tau_{xz}} = -dzdy \left\{ \left(\tau_{xz} + \frac{\partial \tau_{xz}}{\partial x} \frac{dx}{2} \right) \left(w + \frac{\partial w}{\partial x} \frac{dx}{2} \right) - \left(\tau_{xz} - \frac{\partial \tau_{xz}}{\partial x} \frac{dx}{2} \right) \left(w - \frac{\partial w}{\partial x} \frac{dx}{2} \right) \right\}$$

$$\dot{W}_{\tau_{xz}} = -dV \frac{\partial (w \tau_{xz})}{\partial x}$$

$$\dot{W}_{\tau_{xy}} = -dV \frac{\partial (v \tau_{xy})}{\partial x}$$

Performing this operation on the y & z planes also, we get

$$\dot{W} = -dV \left\{ \begin{array}{l} \frac{\partial}{\partial x} (u\sigma_x + v\tau_{xy} + w\tau_{xz}) + \\ \frac{\partial}{\partial y} (u\tau_{yx} + v\sigma_y + w\tau_{yz}) + \\ \frac{\partial}{\partial z} (u\tau_{zx} + v\tau_{zy} + w\sigma_z) \end{array} \right\} - \vec{f} \cdot \vec{v} dV$$

↑
work done by fluid element

↑
body forces by element

aside:

$$v^2 = \sum_{i=1}^3 v_i v_i = u^2 + v^2 + w^2$$

$$\frac{Dv^2}{Dt} = \frac{D}{Dt} (\underline{r} \cdot \underline{v}) = 2 \left(\underline{v} \cdot \frac{D\underline{v}}{Dt} \right)$$

Inserting into $\dot{Q} = \frac{DE}{Dt} + \dot{W}$,

$$\int \frac{DE}{Dt} + \int \underline{v} \cdot \frac{D\underline{v}}{Dt} - \vec{f} \cdot \vec{v} = \nabla \cdot (k \nabla T) + \dot{q}''' + \dot{w}$$

Back to Momentum Eqs. We can write it as

$$\rho \frac{D\underline{v}}{Dt} = \underline{F} + \left[\begin{array}{l} \frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} + \\ \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} + \frac{\partial \tau_{zy}}{\partial z} + \\ \frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_z}{\partial z} \end{array} \right] \quad (\otimes) \text{ Momentum}$$

This expression does not consider pressure gradients yet

$$\rho \left[\bar{v} \cdot \frac{D\bar{v}}{Dt} - \bar{v} \cdot \bar{F} \right] = +v \left(\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} + \frac{\partial \tau_{zy}}{\partial z} \right) + w \left(\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_z}{\partial z} \right) + u \left(\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} \right)$$

rate change of KE = total work done by stresses.

The Energy Eqn provides the rate of change of KE and internal energy. Substituting previous eqn into the Energy eqn yields expression for internal energy only.

$$\rho \frac{De}{Dt} = \nabla \cdot (k \nabla T) + \left\{ \begin{array}{l} \sigma_x \frac{\partial u}{\partial x} + \tau_{xy} \frac{\partial v}{\partial x} + \tau_{xz} \frac{\partial u}{\partial z} + \\ \tau_{xy} \frac{\partial u}{\partial y} + \sigma_y \frac{\partial v}{\partial y} + \tau_{yz} \frac{\partial v}{\partial z} + \\ \tau_{xz} \frac{\partial u}{\partial z} + \tau_{yz} \frac{\partial v}{\partial y} + \sigma_z \frac{\partial w}{\partial z} \end{array} \right\} + \dot{q}''' \quad \text{Energy}$$

With Stokes Assumption ^{heat transfer}

$$\rho \frac{De}{Dt} = \nabla \cdot (k \nabla T) - \underbrace{p \nabla \cdot \bar{v}}_{\substack{\uparrow \\ \text{rate of} \\ \text{compression}}} + \underbrace{\mu \Phi}_{\substack{\uparrow \\ \text{dissipation}}} + \dot{q}''' \quad \text{heat generation}$$

$\nabla \cdot \bar{v} = \frac{1}{\delta T} \frac{\partial \delta V}{\partial x}$

The dissipation function is

$$\Phi = \left\{ \begin{array}{l} 2 \left\{ \left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial y} \right)^2 + \left(\frac{\partial w}{\partial z} \right)^2 \right\} \\ + \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right)^2 + \left(\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z} \right)^2 + \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right)^2 \\ - \frac{2}{3} (\nabla \cdot \bar{v})^2 \end{array} \right\}$$

Unknown variables: $u, v, w, \rho, p, \mu, e, k, T$

We have 1 continuity
3 momentum $\Rightarrow 5$ eqns.
1 energy

Once we know the fluid with thermodynamic properties ρ, μ, e, k , we now have 5 unknowns with 5 eqns $\Rightarrow$ closed problem.

We will assume local thermodynamic properties

The internal energy is a function of say

$$e(p, T)$$

$$e(p, \rho)$$

$$e(\rho, T)$$

For liquids, low dependency on $p \Rightarrow \frac{De}{Dt} \approx C \frac{DT}{Dt}$

$$C_p = \left. \frac{\partial h}{\partial T} \right|_p$$

$$C_v = \left. \frac{\partial h}{\partial T} \right|_p$$

with $h(T, p)$

$$e(T, p)$$

chemical reaction $\Rightarrow e(p, T, n_i, A_0)$

Let $e = h - \frac{p}{\rho}$, a simple fluid (2 coor, no reaction)

$$de = dh - \frac{dp}{\rho} + \frac{p}{\rho^2} d\rho$$

$$\frac{D\rho}{Dt} = \frac{Dh}{Dt} - \frac{1}{\rho} \frac{Dp}{Dt} + \frac{p}{\rho^2} \frac{D\rho}{Dt} \quad -\rho \nabla \cdot \mathbf{v} \text{ (Continuity)}$$

Inserting into Energy eqn,

$$\rho \frac{Dh}{Dt} = \nabla \cdot (k \nabla T) - \frac{1}{\rho} \frac{Dp}{Dt} + \mu \Phi + q'''$$

For a perfect gas ($e(T)$ & $h(T)$ only),

$$Dh = c_p DT$$

$$\frac{Dp}{Dt} \approx 0 \quad \text{assumed}$$

We can show from Thermodynamic Considerations
(will not prove):

$$dh = c_p dT + \frac{1}{\rho} (1 - \beta T) dp$$

where

$$c_p = \left. \frac{\partial h}{\partial T} \right|_p, \quad \beta = -\frac{1}{\rho} \left. \frac{\partial \rho}{\partial T} \right|_p$$

↑
const.
pressure

The energy eqn. becomes:

$$\boxed{\rho c_p \frac{DT}{Dt} = \nabla \cdot (k \nabla T) + \mu \Phi + q''' + \beta T \frac{Dp}{Dt}}$$

with unknowns u, v, w, p, T .

Note: These derivations consider continuous differential changes. Avoid interfaces!! $\Rightarrow$ Not continuous.

for liquids, $\beta = 10^{-6} = -\frac{1}{\rho} \frac{\partial \rho}{\partial T}|_p \approx 0$.

$$C_p \approx C_v$$

for gases at low speed, $\beta T = 1$

$$C_p = 1.4 C_v \text{ (or other way around)}$$

$\Rightarrow 40\% \text{ error.}$

for incompressibility, $\nabla \cdot \mathbf{v} = 0$
or constant ρ

Mass Diffusion

if We look at binary fluids, $i=2$, and so on...

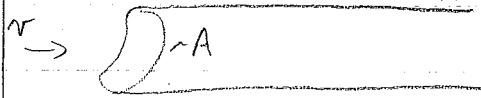
Each fluid has its own properties, where i is the species

$$\frac{\rho_i}{\rho}$$

The rate of flow is

$$\rho_i \vec{v}_i \cdot \hat{n}$$

Laminar HT in Ducts



$$\dot{m} = \int_A \rho v dA \equiv \rho_{\text{avg}} v_{\text{avg}} A$$

We will look constant ρ cases,

$$\int_A v dA = v_{\text{avg}} A$$

The energy in the duct is

$$\dot{E} = \int_A \rho c_p T v dA \equiv \rho_{\text{avg}} c_{p,\text{avg}} v_{\text{avg}} A T_m$$

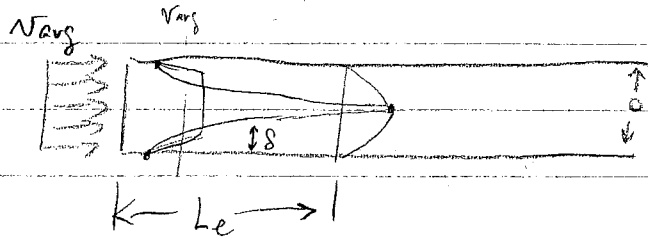
Taking constant ρ & c_p .

T_m is called the mixing-cup temperature

Define heat flux q_w as $\oplus$ wall flux into fluid

$$q_w \equiv h (T_w - T_m)$$

Entrance for circular duct



$$Re = \frac{V_{avg} D}{\nu} \quad \text{high } V, \text{ lower } L_e$$

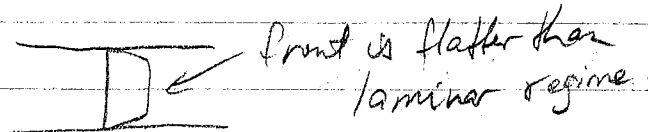
for laminar flows, $\frac{L_e}{D} \approx \frac{Re}{20}$

for turbulent flows, $\frac{L_e}{D} \approx 40-100$, $Re \geq 2200$

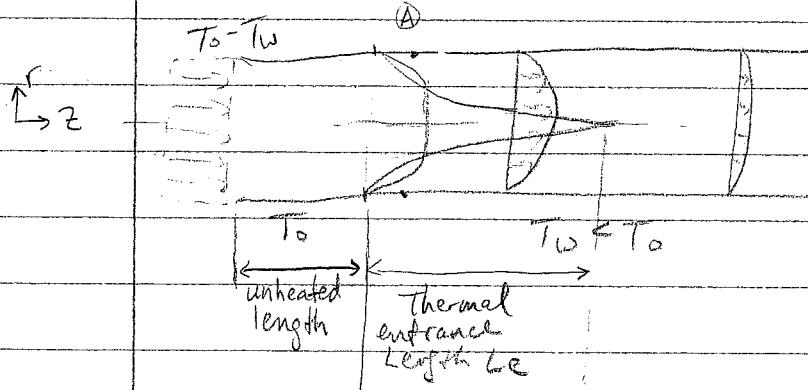
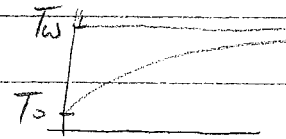
higher Re , shorter L_e

We can do analysis with parabola ($L \geq L_e$) and
slug flow ($L = 0$), but intermediate regions
solved computationally

Turbulent Profile:



Thermal Entry



At A, fluid inside warms up to conserve energy.

At $L > L_e$, h is constant.

Since T is function of r & z only,

$$\frac{T_w - T(r, z)}{T_w - T_{in}} = \psi(r)$$

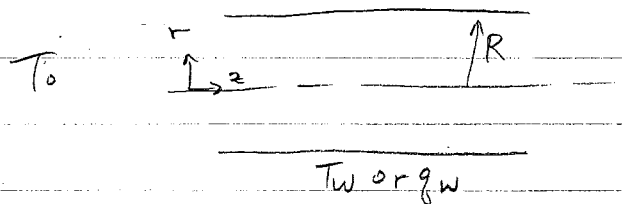
Velocity can be fully developed if

$$L_{unheated} > L_{e \text{ velocity}}$$

Velocity can be assumed slug  if we satisfy two conditions

- 1) $L_{unheated}$ is very short or zero.
- 2) $\frac{L}{\alpha} = P_r \rightarrow 0$ means that temp develops much faster than velocity.

Circular Duct



assumptions:

axisymmetry

steady

laminar

const. prop.

$$\frac{1}{r} \frac{\partial(rv_r)}{\partial r} + \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{\partial v_z}{\partial z} = 0$$

$v(r, z)$



fully developed profile $v_z = v(r)$
 $T(r, z)$

energy eqn:

$$v \frac{\partial T}{\partial z} = \alpha \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) + \frac{\partial^2 T}{\partial z^2} \right] + \frac{\mu}{\rho c_p} \left(\frac{\partial v}{\partial r} \right)^2$$

BC: 1) $T(r, 0) = T_0$

2) $T(0, z)$ is finite needed for $\left(\frac{1}{r} \frac{\partial T}{\partial r} \right)$

3) $T(R, z) = T_w$ or $k \frac{\partial T}{\partial r} \big|_{r=R} = q_w$

Nondimensionalizing the system using scales:

temp: ΔT

radial dir: R

z-dir: L

velocity: V_{avg}

$$\theta = \frac{T - T_r}{\Delta T} \text{ where } T_r = T_0 \text{ or } T_w$$

$$r' = \frac{r}{R}$$

$$z' = \frac{z}{L}$$

$$v' = \frac{v}{V_{avg}}$$

Inserting into eqn,

$$\frac{V_{avg}}{L} \Delta T v' \frac{\partial \theta}{\partial z'} = \alpha \left[\frac{\Delta T}{R^2} \frac{1}{r'} \frac{\partial}{\partial r} \left(r' \frac{\partial \theta}{\partial r} \right) + \frac{\Delta T}{L^2} \frac{\partial^2 \theta}{\partial z'^2} \right] + \frac{\mu}{\rho c_p} \frac{V_{avg}^2}{R^2} \left(\frac{\partial v'}{\partial r'} \right)^2$$

$$1) \theta(r', 0) = \theta_0, \quad \theta_0 = \frac{T_0 - T_r}{\Delta T}$$

$$2) \theta(0, z') \text{ is finite}$$

$$3) \theta(1, z') = \theta_w, \quad \theta_w = \frac{T_w - T_r}{\Delta T}$$

$$\text{or } \frac{\partial \theta}{\partial r'} \bigg|_{r'=1} = \frac{q_w R}{k \Delta T}$$

For constant wall temp, $\Delta T = T_w - T_0$

$$T_r = T_0$$

we get $\theta_0 = 0$ & $\theta_w = 1$.

For constant wall flux, $T_r = T_0$

$$\Delta T = \frac{q_w R}{k}$$

we get $\theta_0 = 0$ & $\theta_w = \frac{T_w - T_0}{\Delta T}$

Simplifying diff eqn,

$$v' \frac{\partial \theta}{\partial z'} = \left(\frac{\alpha L}{\nu_{avg} R^2} \right) \frac{1}{r'} \frac{\partial}{\partial r'} \left(r' \frac{\partial \theta}{\partial r'} \right) + \frac{\alpha}{L \nu_{avg}} \frac{\partial^2 \theta}{\partial z'^2} + \frac{u \nu_{avg} L}{\Delta T_f c_p R^2} \left(\frac{\partial v'}{\partial r'} \right)^2$$

$$\frac{\alpha L}{\nu_{avg} R^2} = 1 \Rightarrow L = \frac{\nu_{avg} R^2}{\alpha}$$

$$\frac{\alpha}{L \nu_{avg}} = \frac{\alpha^2}{\nu_{avg}^2 R^2} = \left(\frac{\alpha}{\nu_{avg} D} \right)^2 4 = 4 \left(\frac{\nu_{avg} D}{\nu} \frac{\nu}{\alpha} \right)^{-2} \quad \text{Prandtl } Pr = \frac{\nu}{\alpha}$$

$$= 4 (Re Pr)^{-2} = \frac{4}{Pe^2}, \quad \text{Peclet } Pe = \frac{D v_{avg}}{\alpha}$$

$$\frac{u \nu_{avg} L}{\Delta T_f c_p R^2} = \left(\frac{\nu_{avg}}{c_p \Delta T} \right) Pr = E Pr, \quad \text{Eckert Number } E = \frac{\nu_{avg}^2}{c_p \Delta T}$$

The eqn becomes

θ -convection

θ -conduction

$$v' \frac{\partial \theta}{\partial z'} = \frac{1}{r'} \frac{\partial}{\partial r'} \left(r' \frac{\partial \theta}{\partial r'} \right) + \frac{4}{Pe^2} \frac{\partial^2 \theta}{\partial z'^2} + Pr E \left(\frac{\partial v'}{\partial r'} \right)^2$$

small

If axial conduction is large, Pe large, $\frac{4}{Pe^2} \frac{\partial^2 \theta}{\partial z'^2} \rightarrow 0$.

The $\frac{\partial^2 \theta}{\partial z'^2}$ term makes the problem elliptic. We need BC on both ends. Can't solve problem analytically with w pipe.

Assume low velocity $E \approx 0$, $Pr E \left(\frac{\partial v'}{\partial r'} \right)^2 \rightarrow 0$.

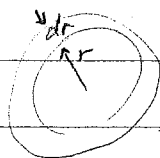
Suppose we solved the problem

$$r' \frac{\partial \theta}{\partial z'} = \frac{1}{r'} \frac{\partial}{\partial r'} (r' \frac{\partial \theta}{\partial r'})$$

for $T(r, z)$ or $\theta(r', z')$

The mixing cup temp is

$$T_m = \frac{2\pi}{\pi R^2} \int_0^R \frac{v}{v_{avg}} T r dr$$



Recall: $\theta = \frac{T - T_r}{\Delta T} \Rightarrow T = T_r + \theta \Delta T$

$$T_m = \frac{2}{R^2} \int_0^R v' (T_r + \theta \Delta T) R^2 r' dr'$$

$$T_m = \Delta T \int_0^R v' \theta r' dr' + 2T_r \int_0^R v' r' dr'$$

Recall: $v_{avg} = \frac{1}{\pi R^2} \int_0^R v 2\pi r dr$

Non-dim: $1 = \frac{2}{R^2} \int_0^R v' R^2 r' dr'$

$$1 = 2 \int_0^1 v' r' dr'$$

$$v' = \frac{v}{v_{avg}}, \quad r' = \frac{r}{R}, \quad dr' = \frac{dr}{R}$$

Insert into eqn,

$$T_m = \Delta T \int_0^1 v' \theta r' dr' + T_r$$

$$\theta_m = \frac{T_m - T_r}{\Delta T} = \int_0^1 v' \theta r' dr'$$

Now, look at heat flux.

$$q_w = k \frac{\partial T}{\partial r} \Big|_{r=R} = h(T_w - T_m)$$

Scaling eqn,

$$q_w = k \frac{\Delta T}{R} \frac{\partial \theta}{\partial r'} \Big|_{r'=1} = h(\theta_w - \theta_m) \Delta T \quad \text{where} \quad \theta_w = \frac{T_w - T_r}{\Delta T} \quad \theta_m = \frac{T_m - T_r}{\Delta T}$$

Define the Nusselt Number

$$Nu = \frac{hD}{k} = \frac{h \Delta T \leftarrow \text{convection}}{k \frac{\Delta T}{D} \leftarrow \text{conduction}}$$

$$Nu = \frac{2 \frac{\partial \theta}{\partial r'} \Big|_{r'=1}}{\theta_w - \theta_m}$$

We are interested in the eqn

$$r' \frac{\partial \theta}{\partial z'} = \frac{1}{r'} \frac{\partial}{\partial r'} \left(r' \frac{\partial \theta}{\partial r'} \right)$$

$$\theta(r', 0) = \theta_0, \quad \theta_0 = \frac{T_0 - T_r}{\Delta T}, \quad \text{Since } \theta_0 \text{ for all } r' \text{ at } z=0, \theta_m = \theta_0$$

Integrating

$$\int_0^1 r' v' \frac{\partial \theta}{\partial z'} dr' = r' \frac{\partial \theta}{\partial r'} \Big|_0^1$$

Since $v'(r)$ only

$$\frac{\partial}{\partial z'} \left[\int_0^{z'} r r' \theta dr' \right] = \frac{\partial \theta}{\partial r'} \bigg|_{r'=1} - 0$$

$$\frac{1}{2} \frac{d\theta_m}{dz'} = \frac{\partial \theta}{\partial r'} \bigg|_{r'=1}$$

$$\frac{1}{2} \frac{d\theta_m}{dz'} = \frac{\theta_w - \theta_m}{2} Nu \quad \text{from Nu eqn.} \quad \frac{\partial \theta}{\partial r'} \bigg|_{r'=1} = \frac{Nu}{2} (\theta_w - \theta_m)$$

$$\frac{d\theta_m}{dz'} = Nu(\theta_w - \theta_m)$$

If θ_w is prescribed using q_w , we are done!

If θ_w is prescribed using T_w as constant, θ_w is constant.

Nu is still $Nu(z)$ because $h(z)$.

$$\frac{d\theta_m}{\theta_m - \theta_w} = Nu dz'$$

Integrate,

$$\ln \frac{\theta_m - \theta_w}{\theta_{m,0} - \theta_w} = - \int_0^{z'} Nu dz'$$

We define an average Nu over $0 < z < z'$ as

$$- \int_0^{z'} Nu dz' \equiv -\bar{Nu} z'$$